

Part 1: The Big Picture

Why We Need Hadron Models, Hadronization, and Jets

PHYS 7502 – Final Reading Assignment

1 Where We Are in the Story

Up to this point in the course, you’ve been using Griffiths’ *quark model* (Chapters 2, 4, 5) to understand what hadrons *are*: mesons are $q\bar{q}$ bound states, baryons are qqq states, and the quantum numbers of the hadrons we observe follow from the flavors and spins of their “valence” quarks. That picture gets you a long way – it explains the Eightfold Way, it correctly predicts the existence of the Ω^- , it gets meson and baryon magnetic moments roughly right, and so on.

But if you’ve been paying attention, you know it’s not the whole story.

A Candid Assessment

The Gell-Mann quark model in Griffiths Ch. 2/4/5 is *not* a holistic model of hadronic structure – it was never meant to be. It captures only the **leading contributions** (the valence quarks) to the expected quantum numbers of known hadrons, and as you saw with the **proton spin problem**, it doesn’t even do that perfectly. (Recall: when experimentalists measured how much of the proton’s spin actually resides in the quarks, they found only about 30%, not 100% as the valence quark model predicts.)

So what’s the *real* structure of a hadron? What happens when a high-energy process produces a quark – how does that bare, colored parton turn into the colorless hadrons we actually detect? These are the questions this final reading tackles.

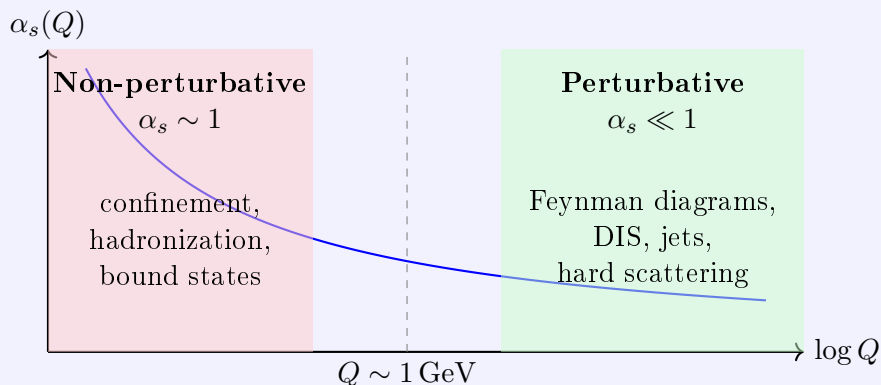
2 The Central Problem: Hadronization

2.1 What we can calculate, and what we can’t

You’ve learned that QCD has the marvelous property of **asymptotic freedom**: at high momentum transfer Q , the strong coupling $\alpha_s(Q^2)$ becomes small, and perturbation theory works. This is what lets you compute things like $e^+e^- \rightarrow q\bar{q}$ cross sections using Feynman diagrams and Casimir’s trick, exactly as you did for Compton scattering and electron-muon scattering.

But at *low* Q – below about 1 GeV – the coupling becomes $\mathcal{O}(1)$, the perturbation series stops converging, and everything we know how to do breaks down. This is the regime where quarks actually get confined into hadrons. You cannot compute hadronization using Feynman diagrams. Full stop.

The Two Regimes of QCD



The vertical dashed line is the rough dividing line between the two regimes. Hadronization happens *entirely on the left side* – which is precisely why it’s so hard.

2.2 Lattice QCD: the only first-principles answer

The one technique that *can* in principle solve the non-perturbative problem from first principles is **Lattice QCD**: you discretize spacetime into a grid, put the quark and gluon fields on the grid, and numerically integrate the QCD path integral. With enough computer time, this gives you the “full” answer – and lattice QCD has successfully predicted the proton mass to within a few percent, the hadron spectrum, and many other static quantities.

But lattice QCD has serious limitations in practice:

- **Grid size.** Your lattice has to be big enough to contain the physics you care about, but fine enough to resolve short-distance dynamics. This costs computer time – a lot of it. A single calculation can take months on specialized supercomputers.
- **“Steady-state” limitation.** Lattice works best for equilibrium quantities (masses, decay constants, form factors). It struggles with *time-dependent* processes – and hadronization is fundamentally a dynamical, time-dependent process.
- **Sign problem.** At finite baryon density (relevant for heavy-ion collisions and neutron stars), lattice QCD hits the infamous *sign problem* and essentially fails.

The Takeaway

Lattice QCD is our best hope for a fundamental theory of hadronization, but we can’t use it yet for most things we care about. So we build **models** – simplified frameworks that capture the essential physics and that we tune to data.

3 A Taxonomy of Hadron Models

The literature contains many hadron models, and they come in quite different flavors. It helps to sort them into categories before we dive into the details.

3.1 Structural / dynamical models

These try to describe the hadron as a bound state with actual equations of motion – some non-perturbative mechanism that holds the quarks and gluons together.

- **The MIT Bag Model** (Part 8). Quarks are treated as free particles inside a “bag” that confines them via an external pressure B . Crude, but captures the essence of confinement and even predicts the phase transition to quark-gluon plasma.
- **String / Lund model** (Parts 5–6). The color field between a q and $\bar{q}$ is squeezed into a narrow tube with linear potential $V(r) = \kappa r$. The string breaks by spontaneous $q\bar{q}$ pair production, producing hadrons. This is what PYTHIA uses.
- **Cluster model**. Color-singlet $q\bar{q}$ combinations produced after the parton shower are grouped into “clusters” that decay isotropically into hadrons. This is what the retired HERWIG generator used.

3.2 Classification models

These don’t try to describe dynamics – they just organize the observed hadron states and their scattering.

- **Eightfold Way** (the original – you already know this). Organizes mesons and baryons into $SU(3)_{\text{flavor}}$ multiplets based on light-quark content.
- **Reggeon model** (Part 7). Organizes hadrons by their *Regge trajectories* – straight lines in (J, M^2) space that connect states with the same internal quantum numbers but different spins. This was developed *before* the quark model and is still useful for soft scattering.

3.3 Guiding principles

These aren’t full models – they’re heuristics that constrain more detailed models.

- **Parton-hadron duality**. In its simplest form: when converting partons to hadrons, you can often just “relabel” partons as hadrons, preserving the 3-vector or 4-vector momentum. Crude, but surprisingly effective.
- **Factorization**. Cross sections for processes involving hadrons can be written as a convolution of a perturbative partonic cross section with non-perturbative functions (PDFs, fragmentation functions). You’ve met this in DIS; Part 4 extends it to fragmentation.

3.4 Statistical / thermal models

A relative newcomer to the scene. In high-multiplicity environments (heavy-ion collisions, sometimes even pp), you can describe hadron production using thermal-equilibrium statistical mechanics – without ever mentioning the underlying microscopic dynamics. This works surprisingly well.

From Prof. Frantz

QGP formation has developed into its own “higher level” model of hadron formation. Along with similar statistical/thermal approaches, it may apply even to simpler $p+p$ collisions – not just the rare events where RHI physicists think QGP might actually form. However, these descriptions never include the microscopic-level processes – they are only statements about

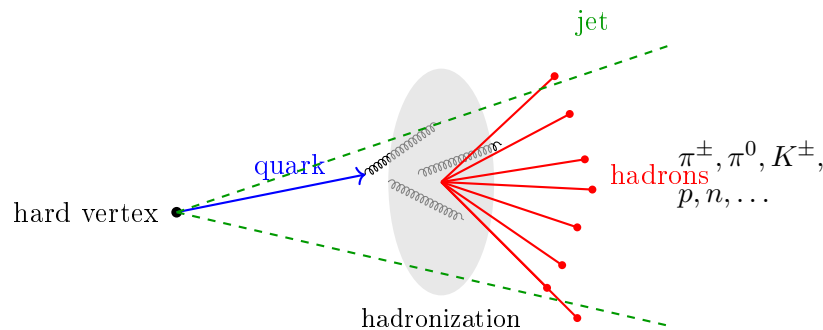
the thermal/statistical properties of the many-body interactions in many-particle systems.

4 Why Jets Are Central

Of all the tools we have for studying hadrons and hadronization, **jets** are arguably the most important. Here’s why.

4.1 Jets are how we “see” quarks and gluons

When a hard scattering produces a bare quark or gluon, that parton can’t just fly out of the detector – it has color charge, and color charges don’t propagate freely. What you see in the detector is a **jet**: a collimated spray of hadrons roughly following the direction of the original parton.



The quark radiates gluons (the perturbative *parton shower*), those gluons can split into more partons, and eventually – at the scale where $\alpha_s \sim 1$ – everything non-perturbatively rearranges itself into color-singlet hadrons. What reaches your detector is mostly pions, with some kaons, baryons, and photons from π^0 decays mixed in.

Why Jets Matter

A jet is the experimental shadow of a parton. If you can reconstruct the jet well, you can recover (approximately) the parton’s kinematics – its energy, its direction, even clues about whether it was a quark or a gluon. This makes jets the primary link between the theoretical language of QCD (quarks and gluons) and what detectors actually measure (hadrons).

4.2 Jets as QCD objects in their own right

There’s another reason to care about jets: at high energy, a jet is itself almost a “hadron-like” object. It has a mass, a flavor content, a substructure, even a notion of “color charge”. Modern jet physics treats jets as QCD objects and studies their internal structure (substructure) to tag heavy particles, distinguish quark from gluon jets, and search for new physics.

5 What This Reading Covers

Here’s the roadmap for the rest of this series of PDFs:

The Eight-Part Plan

- Part 1 (this document):** The big picture – why we need hadron models, why lattice QCD alone isn’t enough, and why jets are central.
- Part 2:** Jets at the LHC – experimental definition, jet algorithms (cone, k_T , anti- k_T), infrared and collinear safety, boosted jets and substructure.
- Part 3:** Event generators – PYTHIA, HERWIG, SHERPA, HIJING. The Monte Carlo tools that translate theory into simulated events.
- Part 4:** Fragmentation functions – the non-perturbative object that parametrizes parton-to-hadron conversion, and how it fits into factorization theorems.
- Part 5:** The Schwinger mechanism – pair production in strong fields, and how this motivates the color flux tube picture of confinement.
- Part 6:** The classical string model and the Lund model – the dynamical picture behind PYTHIA’s hadronization.
- Part 7:** The Dual Parton Model, Regge trajectories, and the Pomeron – the framework for soft/diffractive scattering.
- Part 8:** The MIT Bag Model and the quark-gluon plasma transition – a different (and complementary) picture of hadron structure that also gets you to QGP.

6 A Unifying Thread: The String Tension κ

Before you dive in, here’s a preview of a theme that will recur across almost every one of these parts. There’s a single parameter that keeps showing up everywhere:

The String Tension

The **color string tension** – the energy per unit length stored in the color field between a separated $q\bar{q}$ pair – is

$$\kappa \approx 1 \text{ GeV/fm} \quad (1)$$

This single number appears in:

- The *Schwinger mechanism* as the strength of the color electric field (Part 5).
- The *Lund string model* as the string tension (Part 6).
- The *Regge slope* $\alpha' = 1/(2\pi\kappa) \approx 1 \text{ GeV}^{-2}$ (Part 7).
- The *linear confinement potential* $V(r) = \kappa r$ (Parts 6, 8).
- The *bag pressure* $B^{1/4} \sim 206 \text{ MeV}$, which indirectly reflects the same scale (Part 8).

One parameter, multiple manifestations. When you see κ appear in a new context, know that it’s the same κ – the same underlying physics of confinement showing itself in different ways.

7 Summary of Part 1

The Main Points

1. The valence quark model is a useful but incomplete picture of hadrons. It fails on things like the proton spin problem because it only captures leading contributions.
2. Hadronization happens at low Q , where $\alpha_s \sim 1$ and perturbation theory doesn't work. This is why it's hard.
3. Lattice QCD is the only first-principles tool for non-perturbative QCD, but it has practical limits (especially for time-dependent processes).
4. A range of *models* fills the gap: structural models (bag, string, cluster), classification models (Regge, Eightfold Way), guiding principles (duality, factorization), and statistical approaches (thermal models, QGP).
5. **Jets** are how we see colored partons in the detector – they are the central experimental object of this reading.
6. The color string tension $\kappa \approx 1 \text{ GeV/fm}$ is a unifying thread across nearly every model we'll discuss.

The rest of the series builds on these foundations. Onward to Part 2: jets themselves.